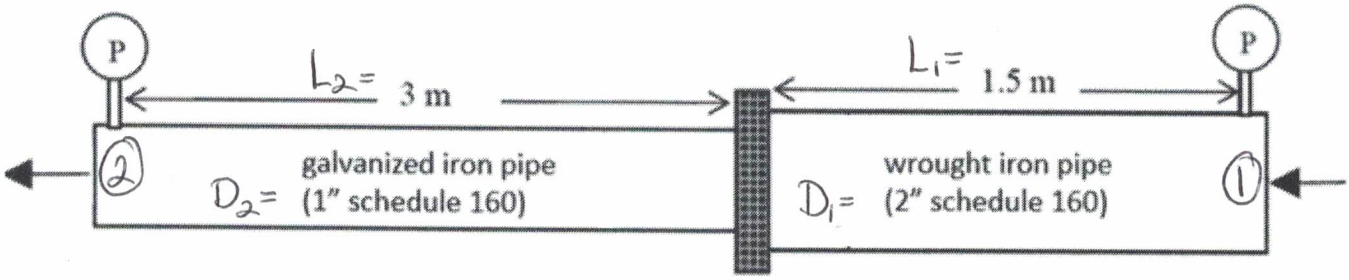
Tutorial 9

Carbon tetrachloride at ambient temperature (viscosity = 0.001 Pa·s) is flowing through the horizontal piping system as shown below (not drawn to scale). A contraction fitting (also known as a reducing bushing) is used to connect the larger wrought iron pipe (2" schedule 160) to the smaller galvanized iron pipe (1" schedule 160).



a. Starting from a steady-state mass balance, write the appropriate equation that relates the average velocities in the two pipes.

$$\dot{m}_1 = \dot{m}_2$$
$$\cancel{\rho} Q_1 = \cancel{\rho} Q_2$$
$$V_1 A_1 = V_2 A_2$$
$$V_2 = \frac{A_1}{A_2} \cdot V_1$$

b. Write the simplified form of the Bernoulli equation for the condition where reference points 1 & 2 are chosen to be at the centreline of the piping system where the two pressure gauges are located.

$Z_1 = Z_2$ and so the Bernoulli equation is:

$$\frac{P_1}{\rho} + \alpha_1 \frac{V_1^2}{2} = \frac{P_2}{\rho} + \alpha_2 \frac{V_2^2}{2} + f_1 \frac{L_1}{D_1} \frac{V_1^2}{2} + f_2 \frac{L_2}{D_2} \cdot \frac{V_2^2}{2} + K_{contraction} \frac{V_2^2}{2}$$

c. If the upstream pressure gauge reads 192 kPa and the downstream pressure gauge reads 20 kPa, determine the volumetric flow rate of carbon tetrachloride (m^3/s) in the piping system.

Combine equations from a) and b) to get in terms of just V_1:

$$\frac{P_1}{\rho} + \alpha_1 \frac{V_1^2}{2} = \frac{P_2}{\rho} + \alpha_2 \frac{1}{2}\left(\frac{A_1}{A_2} V_1\right)^2 + f_1 \frac{L_1}{D_1} \cdot \frac{V_1^2}{2} + f_2 \frac{L_2}{D_2} \cdot \frac{1}{2}\left(\frac{A_1}{A_2} V_1\right)^2 + K_{contraction} \frac{1}{2}\left(\frac{A_1}{A_2} V_1\right)^2$$

now rearrange to solve for V_1:

$$\frac{2(P_1 - P_2)}{\rho} = V_1^2 \left[\alpha_2 \left(\frac{A_1}{A_2}\right)^2 + f_1 \frac{L_1}{D_1} + f_2 \frac{L_2}{D_2}\left(\frac{A_1}{A_2}\right)^2 + K_{contraction} - \alpha_1\right]$$

$$V_1 = \sqrt{\frac{2(P_1 - P_2)}{\rho\left[\alpha_2 \left(\frac{A_1}{A_2}\right)^2 + f_1 \frac{L_1}{D_1} + f_2 \frac{L_2}{D_2}\left(\frac{A_1}{A_2}\right)^2 + K_{contraction} - \alpha_1\right]}}$$